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MIT School of Computing

Name of Examination: Term Assessment – I Examination Aug-2024

Programme: B. Tech. Year: LY Academic Year: 2024-25

Branch: Computer Science & Engineering Semester: VII

Subject Code: 21BTCS030 Subject Name: High Performance Computing

Date: 26/08/2024 Time: 11:30 am to 12:30 pm Max. Marks: 20

Instructions: -

1. Attempt the Que 1 OR Que 2 and Que 3 OR Que 4.
2. Neat diagrams must be drawn, wherever necessary.
3. Use of slide rule, Mollier chart, electronic scientific calculator and steam tables are allowed.
4. Figures to right indicate the marks allotted to the questions.

Q1. Attempt the following questions

[10]

UNIT-1

- (i) A scientific simulation is distributed across a cluster of computers, each executing different parts of the simulation concurrently on different data sets. Which architecture as per Flynn's Classification can be used to implement the same. Draw the architecture and explain the reason for choosing the same.
- (ii) Consider 10% of the code is serial in nature and we use a system with 40 cores. Calculate the absolute and maximum possible speedup that can be achieved.
- b. Explain why the performance of a hardware multithreaded processing core might degrade if it had large caches and it ran many threads.

OR

Q2. Attempt the following questions

[10]

UNIT-1

- Define cache hit ratio. Consider the following set of serial instructions for performing dot product:

S1: Load R1, [R2]

S2: Load R3, [R4]

S3: Madd R5, R1, R3

Say, S1 and S2 takes 100ns each for execution and we avoid execution time for S3 for the time being.

What would be the approximate cache hit ratio in this case. Calculate the average memory access time for the above set of statements for a single iteration.

- b. Apply Foster's method to an embarrassingly parallel problem— vector addition. What are the tasks? What are the communications among the tasks? How will you aggregate the tasks? How will you map composite tasks to processes/threads?

Q3. Attempt the following questions

[10]

UNIT-2

- a. Write an OpenMP program to generate the following sequence for n terms: 0,1,2,3,6, 12,.....upto n terms.
- b. Consider the loop
- ```
a [0] = 0;
for (i = 1; i < n ; i ++)
 a [i] = a [i-1] + i ;
```

There's clearly a loop-carried dependence, as the value of  $a[i]$  can't be computed without the value of  $a[i-1]$ . Can you see a way to eliminate this dependence and parallelize the loop?

**OR**

**Q4. Attempt the following questions**

**[10]**

**UNIT-2**

- a. Implement a simple parallel program in C that initializes an array of integers with the sequence 1,12,123,1234, 12345,....upto n terms and calculates the sum of its elements concurrently using OpenMP. Identify potential issues with concurrent updates to the total\_sum variable in the parallelized version of the program. Modify the program to use OpenMP's critical directive to ensure synchronization.
- b. Use OpenMP to implement a producer consumer program in which some of the threads are producers and others are consumers. The producers read text from a collection of files, one per producer. They insert lines of text into a single shared queue. The consumers take the lines of text and tokenize them. Tokens are "words" separated by white space. When a consumer finds a token, it writes it to stdout.

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